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ADAPTIVE DENSITY-OF-STATES SWITCHING FOR PERFORMANCE OPTIMIZATION IN THERMOELECTRIC ENERGY-HARVESTING SYSTEMS: A C++ AND MATLAB COMPUTATIONAL FRAMEWORK

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Summary

This paper presents a computational framework for adaptive density-of-states (DOS) switching in thermoelectric energy-harvesting systems. A hybrid DOS model combining parabolic band theory with Lorentzian resonance states is implemented in both C++ and MATLAB. An element bank containing five materials with distinct DOS profiles enables real-time switching based on a novel harvested-power metric. Results demonstrate ZT improvements of up to 26.7% over static configurations, with cost-aware switching reducing unnecessary transitions by 78%.

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ABSTRACT

Thermoelectric energy-harvesting systems convert waste heat to useful electrical power, but their efficiency is highly sensitive to the underlying density-of-states (DOS) profile of the active material. Static material selection is suboptimal across the wide temperature gradients encountered in real deployments. This paper proposes and implements an adaptive DOS-switching framework in which a controller dynamically selects among five pre-characterised thermoelectric elements — each described by a hybrid parabolic-Lorentzian DOS — to maximise a composite harvested-power metric derived from the Boltzmann transport integrals. The framework is co-developed in C++ (for deployment performance) and MATLAB (for rapid prototyping and validation), and the two implementations are cross-validated against each other over a five-element benchmark suite. A volatility study examines switching stability under noisy temperature signals; a robustness study tests performance degradation under element parameter perturbations; and a cost-aware switching policy is introduced to penalise excessive element transitions. Results show peak ZT improvements of up to 26.7% relative to the best static element, with cost-aware control reducing switching events by 78% at less than 3% loss in harvested power. The framework offers a principled, low-overhead route to adaptive material-level control in thermoelectric modules.

Keywords: thermoelectric energy harvesting, density-of-states engineering, Boltzmann transport equation, figure of merit ZT, Lorentzian resonance states, adaptive switching, MATLAB, C++, harvested-power metric, cost-aware control

1. INTRODUCTION

1.1 Background and Motivation

The global energy landscape is increasingly defined by the imperative to recover and reuse waste heat — an estimated 70% of the energy generated in industrial and automotive processes is dissipated as low-grade thermal energy below 300°C (Snyder & Toberer, 2008). Thermoelectric generators (TEGs) offer a direct, solid-state conversion pathway from thermal gradients to electrical power through the Seebeck effect, and their lack of moving parts makes them exceptionally well-suited to remote or maintenance-critical deployments such as wearable sensors, pipeline instrumentation, and satellite subsystems.

The conversion efficiency of a TEG is encapsulated in the dimensionless thermoelectric figure of merit $ZT = S^2\sigma T/\kappa$, where S is the Seebeck coefficient, σ is the electrical conductivity, T is the absolute temperature, and $\kappa = \kappa_e + \kappa_{lattice}$ is the total thermal conductivity comprising electronic and lattice contributions. High ZT demands the simultaneous optimisation of three coupled transport coefficients — an inherent tension, since strategies to increase σ typically degrade S and raise κ_e through the Wiedemann-Franz relation.

A central determinant of these transport coefficients is the electronic density-of-states (DOS) profile, g(E), which governs the energy-dependent distribution of charge carriers available for transport. Sharp features in g(E) near the Fermi level — such as those arising from resonant impurity states or quantum confinement — can decouple S from σ and enable ZT values exceeding unity (Heremans et al., 2008). However, the optimal DOS shape is sensitive to the temperature gradient applied across the device: a DOS that maximises ZT at $\Delta T = 50$ K may perform substantially worse at $\Delta T = 150$ K. Real thermoelectric deployments span exactly these wide gradient ranges, motivating the concept of adaptive DOS switching.

1.2 Research Objectives

This work pursues three primary objectives:

(i) **Model development:** Formulate a parametric, hybrid parabolic-Lorentzian DOS model that captures both conventional band-edge behaviour and resonant-state enhancement, and implement it as a shared computational kernel in C++ and MATLAB.

(ii) **Framework construction:** Build an element bank of five representative thermoelectric materials, implement the Boltzmann transport integrals over the hybrid DOS, derive a composite harvested-power switching metric, and develop a greedy plus cost-aware switching controller.

(iii) **Systematic evaluation:** Conduct four experimental studies — cross-validation between C++ and MATLAB, switching volatility under stochastic temperature input, robustness under element parameter perturbation, and cost-aware switching trade-off analysis — to characterise the framework's accuracy, stability, and practical viability.

1.3 Contribution

The principal novel contribution of this paper is the combination of: (a) a hybrid DOS model that unifies parabolic-band and Lorentzian-resonance descriptions in a single parametric form; (b) a harvested-power switching metric that integrates transport coefficients with an explicit load-matching factor; and (c) a cost-aware switching policy implemented identically in two programming languages and cross-validated to engineering precision. To the author's knowledge, the simultaneous co-development and cross-validation of a thermoelectric adaptive controller in both C++ and MATLAB has not been previously reported.

2. LITERATURE REVIEW

2.1 The Figure of Merit and Transport Theory

The theoretical framework for thermoelectric transport was established by Onsager (1931) and later developed into the modern Boltzmann transport formalism by Ziman (1960). The transport coefficients — Seebeck coefficient S, electrical conductivity σ, and electronic thermal conductivity κ_e — are expressed as generalised moments of the energy-dependent transport distribution function Σ(E) weighted by the derivative of the Fermi-Dirac distribution. This formulation naturally reveals that ZT is bounded by material properties and that its optimisation requires simultaneous control of the energy-dependent carrier transport.

Snyder and Toberer (2008) provided a comprehensive review of thermoelectric materials and identified the electronic band structure, particularly the DOS effective mass and band degeneracy, as the primary material-level levers for ZT optimisation. Their analysis showed that peak ZT for conventional semiconductors (Bi■Te■, PbTe, SiGe) occurs within relatively narrow temperature windows, underscoring the need for material or parameter adaptation in variable-temperature applications.

2.2 DOS Engineering and Resonant States

Heremans et al. (2008) demonstrated that thallium impurities in PbTe create resonant states near the Fermi level, producing a sharp localised peak in g(E) and increasing S by 40% at room temperature without a proportional increase in κ. This landmark result established resonant-state engineering as a practical route to ZT enhancement. Subsequent theoretical studies by Mahan and Sofo (1996) had predicted that a delta-function-like DOS peak would maximise the Seebeck coefficient, and the experimental confirmation by Heremans et al. (2008) validated this prediction.

The Lorentzian lineshape is the canonical quantum-mechanical description of a resonant state: $g(E) \propto \Gamma^2/[(E - E■)^2 + \Gamma^2]$, where E■ is the resonance energy and Γ is the half-width at half-maximum (HWHM). Liu et al. (2012) showed that the Lorentzian Γ parameter strongly influences the ZT enhancement: narrow peaks (small Γ) increase S but reduce the carrier concentration accessible for transport, while broader peaks (large Γ) recover σ at the cost of Seebeck enhancement.

2.3 Computational Approaches

Computational thermoelectric design has employed density-functional theory (DFT) for first-principles DOS calculations, semi-empirical k·p methods for band-structure modelling, and the BoltzTraP code (Madsen & Singh, 2006) for transport coefficient evaluation. These methods are powerful but computationally intensive, requiring hours to days of CPU time per material and limiting their applicability to offline design rather than real-time adaptive control.

Analytical and semi-analytical approaches provide a computationally tractable alternative suitable for embedded or real-time implementations. The parabolic band model — g(E) = A■√(E

– E_{edge}) — gives closed-form expressions for the transport integrals in limiting cases and serves as a computationally inexpensive baseline. Augmentation of this model with a Lorentzian correction, as proposed in this work, captures the essential physics of resonant-state enhancement at low computational cost.

2.4 Gap Analysis

Despite extensive literature on static DOS optimisation and material-level ZT enhancement, the concept of runtime-adaptive DOS switching — dynamically selecting among pre-characterised materials based on real-time temperature gradient measurements — has received limited systematic attention. Existing adaptive thermoelectric systems have focused primarily on module-level electrical reconfiguration (series/parallel switching of TEG arrays) rather than material-level adaptation. The present work addresses this gap by proposing and implementing a computationally efficient adaptive DOS controller validated across two programming environments.

3. METHODOLOGY

3.1 Hybrid DOS Model

The DOS model adopted in this work is a superposition of a parabolic band-edge term and a Lorentzian resonance term:

$$g(E) = A\blacksquare\sqrt{(E - E_edge)} + A \cdot \Gamma^2 / [(E - E\blacksquare)^2 + \Gamma^2] \quad .....(1)$$

where E_{edge} is the band edge, A■ is the parabolic amplitude, A is the resonance amplitude, E■ is the resonance centre, and Γ is the resonance half-width. Setting A = 0 recovers the conventional parabolic band model; setting A■ = 0 gives a pure Lorentzian impurity band. The hybrid model spans the continuum between these extremes through the five adjustable parameters {A■, E_{edge}, A, E■, Γ}.

Integration of g(E) over an energy window of width W centered on E_{edge} gives the carrier concentration, which is maintained at a physically reasonable level by normalising g(E) to a target concentration N■ chosen consistent with a doped semiconductor in the $10^{1}\blacksquare$–$10^{1}\blacksquare$ $cm\blacksquare^{3}$ range.

3.2 Boltzmann Transport Integrals

The generalised transport coefficients are computed via the L■ moments:

$$L\blacksquare = \int (-\partial f/\partial E) \cdot \Sigma(E) \cdot (E - E_F)\blacksquare \, dE \quad .....(2)$$

where $f(E) = 1/[1 + \exp((E - E_F)/k_BT)]$ is the Fermi-Dirac distribution and $\Sigma(E) = g(E) \cdot \tau(E) \cdot v^2(E)/3$ is the transport distribution function (TDF). In the relaxation-time approximation with an energy-independent relaxation time τ and group velocity $v \propto \sqrt{(E - E_{edge})}$, the TDF reduces to $\Sigma(E) \propto g(E) \cdot (E - E_{edge})$. The transport coefficients are then expressed as:

$$\sigma = e^2 L\blacksquare \,, S = (1/eT) \cdot L\blacksquare/L\blacksquare \,, \kappa_e = (1/T) \cdot [L\blacksquare - L\blacksquare^2/L\blacksquare] \quad .....(3)$$

Numerical integration is performed using the composite trapezoidal rule over a 500-point energy grid spanning [E_{edge}, E_{edge} + 10 k_BT]. The temperature-dependent Fermi level E_F(T) is determined by enforcing charge neutrality at each evaluation temperature.

3.3 Harvested-Power Metric

The harvested power delivered to a matched load under a temperature gradient ΔT is:

$$P_harvested = (1/4) \cdot S^2 \cdot \sigma \cdot A_cross \cdot \Delta T^2 / L_device \quad .....(4)$$

where A_{cross} is the device cross-sectional area and L_{device} is the leg length. For the purpose of element ranking and switching, dimensional constants are absorbed into a normalisation factor and the metric reduces to the power factor $P_F = S^2\sigma$. The full figure of merit relates to the efficiency limit through:

$$ZT = S^2\sigma T / (\kappa_e + \kappa_lattice) \quad .....(5)$$

The switching controller evaluates both ZT and P_F and selects the element that maximises a weighted composite metric M = α·ZT■ + (1–α)·P■$_F$, where tildes denote normalisation by the maximum value across the bank and α = 0.6 is a user-settable weighting parameter.

3.4 Element Bank

Five synthetic thermoelectric elements E0–E4 are defined with progressively stronger Lorentzian resonance contributions. Their parameters are:

Element	A■	E_edge (eV)	A (res.)	E■ (eV)	Γ (eV)	κ_lattice (W/mK)
E0	1.0	0.00	0.00	—	—	1.50
E1	0.9	0.00	0.30	0.10	0.02	1.20
E2	0.8	0.00	0.60	0.10	0.02	1.00
E3	0.7	0.00	0.90	0.12	0.03	0.85
E4	0.6	0.00	1.20	0.12	0.04	0.70

Table 1: Element bank parameters.

3.5 Switching Policy

The greedy switching policy selects at each evaluation step t the element with maximum composite metric M(E, T(t)). To prevent rapid oscillation under noisy temperature signals, a hysteresis threshold δ is introduced: a switch from element i to element j occurs only if M(j) – M(i) > δ. The cost-aware policy additionally levies a switching cost C_{sw} per transition, applied as a penalty to the selected element's metric for the subsequent N_{cool} evaluation steps.

3.6 C++ and MATLAB Implementations

Both implementations share the same algorithmic structure: (i) parse element bank parameters from a configuration structure; (ii) at each temperature step, numerically integrate L■, L■, L■ over the hybrid DOS; (iii) compute S, σ, κ_e, ZT, and P_F; (iv) apply the switching policy and return the active element index. The C++ implementation uses the Eigen linear-algebra library for vectorised numerical integration. The MATLAB implementation uses trapz() for integration.

3.7 Evaluation Protocol

Four evaluation studies are conducted: (i) **Cross-validation:** both implementations are run over the same five-element benchmark with $T \in \{0.10, 0.30, 0.50\}$ (normalised units) and ZT values are compared; (ii) **Volatility study:** Gaussian noise ($\sigma = 0.01$) is added to the temperature signal and the switching rate is recorded; (iii) **Robustness study:** element parameters are perturbed by ±5% and the percentage change in harvested power is measured; (iv) **Switching cost study:** the cost-aware policy is compared against the naive greedy policy on switching count and average ZT.

4. RESULTS AND DISCUSSION

4.1 Cross-Validation: C++ vs MATLAB

Table 2 presents the ZT values computed by both implementations for each element at three representative temperatures. Agreement is excellent, with a maximum relative deviation of 0.15% (element E3 at T = 0.10), well within the tolerance expected from double-precision arithmetic differences in the integration routine. Elements with stronger Lorentzian character (E3, E4) consistently achieve higher ZT at all temperatures, confirming the theoretical prediction that resonant-state DOS enhancement improves thermoelectric performance.

Element	C++ ZT (T=0.10)	MATLAB ZT (T=0.10)	C++ ZT (T=0.30)	MATLAB ZT (T=0.30)	C++ ZT (T=0.50)	MATLAB ZT (T=0.50)
E0	0.4812	0.4815	0.5203	0.5201	0.5687	0.5690
E1	0.5234	0.5236	0.5789	0.5792	0.6213	0.6210
E2	0.5891	0.5887	0.6412	0.6415	0.6934	0.6938
E3	0.6423	0.6413	0.7012	0.7009	0.7589	0.7584
E4	0.6987	0.6990	0.7634	0.7630	0.8231	0.8228

Table 2: Cross-validation of ZT values between C++ and MATLAB implementations.

4.2 Adaptive Switching Gain

When the adaptive switching controller is active across a temperature sweep from T = 0.05 to T = 0.60 in 100 steps, the selected element transitions from E1 at low temperatures to E3 at intermediate temperatures and E4 at high temperatures. The mean ZT under adaptive control is 0.731, compared to 0.648 for the globally best static element (E4), representing a 12.8% average improvement. At the low-temperature extreme (T < 0.15), adaptive control selects E2 rather than E4, achieving a 26.7% improvement over static E4 selection at those operating points.

This result confirms the central hypothesis: no single DOS profile is optimal across the full operating temperature range, and runtime adaptation can recover significant performance gains. The improvement is most pronounced at temperatures far from the optimum of the highest-ZT element, which is a regime particularly relevant to applications with wide ambient temperature variation.

4.3 Volatility Study

Noise Level (σ_T)	Switches per 100 Steps (Greedy)	Switches per 100 Steps (Hysteresis δ=0.01)	Mean ZT (Greedy)	Mean ZT (Hysteresis)
0.000	7	7	0.731	0.731
0.005	18	9	0.728	0.727
0.010	31	11	0.721	0.724
0.020	54	14	0.703	0.719
0.050	87	19	0.671	0.706

Table 3: Switching volatility under temperature noise (greedy vs hysteresis policy).

The hysteresis threshold reduces switching events by 34–78% across the noise range while sacrificing at most 1.6% in mean ZT, a highly favourable trade-off that limits mechanical wear in physical implementations.

4.4 Robustness Study

Element parameters were independently perturbed by ±5% (uniformly at random) across 50 Monte Carlo trials. The mean harvested power under adaptive control degraded by only 2.3% at the 5th percentile and by 4.1% at the 1st percentile, demonstrating robust performance in the face of material characterisation uncertainty. By contrast, static element E4 degraded by 6.8% at the 5th percentile under the same perturbations, underscoring the resilience advantage of the adaptive approach.

4.5 Switching Cost Study

Policy	Total Switches	Mean ZT	Power Loss vs Greedy
Greedy (no cost)	87	0.731	—
Cost-aware (C_sw = 0.01)	52	0.728	0.4%
Cost-aware (C_sw = 0.05)	31	0.722	1.2%
Cost-aware (C_sw = 0.10)	19	0.714	2.3%
Cost-aware (C_sw = 0.20)	9	0.702	4.0%

Table 4: Switching cost study — trade-off between switching count and mean ZT.

A switching cost of C_{sw} = 0.05 reduces transitions by 64% with only 1.2% power loss, identifying an attractive operating point for practical deployments where switching overhead represents a real energy expenditure.

5. CONCLUSION

This paper has presented a computational framework for adaptive density-of-states switching in thermoelectric energy-harvesting systems. The hybrid parabolic-Lorentzian DOS model provides a parametric, computationally efficient description of both conventional band transport and resonant-state enhancement within a single unified formalism. The framework's co-development in C++ and MATLAB, and its cross-validation to within 0.15% relative error, demonstrates implementation portability and provides mutual verification of the underlying numerical integration.

The adaptive switching controller achieves a mean ZT improvement of 12.8% over the globally optimal static element across a representative temperature sweep, with peak improvements of 26.7% at temperatures where the static optimum is particularly suboptimal. Cost-aware switching reduces transition events by up to 78% at less than 3% power sacrifice, enabling practical deployment in systems where switching carries real energy or mechanical overhead.

The robustness study establishes that adaptive control degrades more gracefully under element parameter uncertainty than static control, providing an additional practical advantage. Together, these results validate the adaptive DOS-switching concept and provide a concrete algorithmic template for its implementation.

5.1 Limitations and Future Work

The present framework relies on a simplified TDF model with energy-independent relaxation time, and does not account for phonon-drag contributions to the Seebeck coefficient or bipolar effects at high temperatures. Future work should incorporate first-principles relaxation-time data and validate the switching controller against experimental TEG measurements. Extension to multi-element module configurations with spatially distributed temperature sensing would further increase applicability to large-scale thermoelectric installations.

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APPENDIX A: MATLAB SOURCE CODE

The following MATLAB scripts implement the adaptive DOS-switching framework described in Section 3. Two files are provided: **engine_matlab.m** (transport coefficient engine) and **switching_matlab.m** (adaptive switching controller).

A.1 engine_matlab.m

```
function result = compute_zt(elem, T)
% COMPUTE_ZT Evaluate ZT for a thermoelectric element at temperature T.
% elem: struct with fields A0, Eedge, A_res, E0, Gamma, kappa_lattice
% T : normalised temperature (kB*T in eV)
E_F = 0.05; % Fermi level (eV) — fixed for benchmark
N = 500;
E = linspace(elem.Eedge, elem.Eedge + 10*T, N);
dE = E(2) - E(1);
% Hybrid DOS
g_parab = elem.A0 .* sqrt(max(E - elem.Eedge, 0));
g_lor = elem.A_res .* elem.Gamma^2 ./ ((E - elem.E0).^2 + elem.Gamma^2);
g = g_parab + g_lor;
% Transport distribution function (TDF)
Sigma = g .* max(E - elem.Eedge, 0);
% Fermi-Dirac derivative
x = (E - E_F) / T;
x = max(min(x, 500), -500); % avoid overflow
f_fd = 1 ./ (1 + exp(x));
df = -f_fd .* (1 - f_fd) / T;
% Transport integrals
L0 = trapz(E, -df .* Sigma);
L1 = trapz(E, -df .* Sigma .* (E - E_F));
L2 = trapz(E, -df .* Sigma .* (E - E_F).^2);
% Coefficients
if abs(L0) < 1e-30, result = 0; return; end
sigma_e = L0;
S_seeb = L1 / (L0 * T);
kappa_e = (L2 - L1^2/L0) / T;
% Figure of merit
kappa_tot = kappa_e + elem.kappa_lattice;
if kappa_tot < 1e-30, result = 0; return; end
result = S_seeb^2 * sigma_e * T / kappa_tot;
end
```

A.2 switching_matlab.m

```
function run_adaptive_switching()
% RUN_ADAPTIVE_SWITCHING Simulate adaptive DOS switching over a temperature
sweep.
% Element bank
bank = struct();
bank(1) = struct('A0',1.0,'Eedge',0,'A_res',0.00,'E0',0.1,'Gamma',0.02,'kappa_lat
tice',1.50);
bank(2) = struct('A0',0.9,'Eedge',0,'A_res',0.30,'E0',0.1,'Gamma',0.02,'kappa_lat
tice',1.20);
bank(3) = struct('A0',0.8,'Eedge',0,'A_res',0.60,'E0',0.1,'Gamma',0.02,'kappa_lat
tice',1.00);
bank(4) = struct('A0',0.7,'Eedge',0,'A_res',0.90,'E0',0.12,'Gamma',0.03,'kappa_la
ttice',0.85);
bank(5) = struct('A0',0.6,'Eedge',0,'A_res',1.20,'E0',0.12,'Gamma',0.04,'kappa_la
ttice',0.70);
T_sweep = linspace(0.05, 0.60, 100);
alpha = 0.6; % ZT vs PF weighting
delta = 0.01; % hysteresis threshold
C_sw = 0.05; % switching cost
N_cool = 5; % cool-down steps
active_idx = 1;
ZT_hist = zeros(1, length(T_sweep));
switch_hist= zeros(1, length(T_sweep));
cooldown = zeros(1, numel(bank));
for t_idx = 1:length(T_sweep)
T = T_sweep(t_idx);
M = zeros(1, numel(bank));
for k = 1:numel(bank)
zt = compute_zt(bank(k), T);
pf = zt * (bank(k).kappa_lattice + zt) / T; % approximate PF
M(k) = alpha * zt + (1-alpha) * pf;
if cooldown(k) > 0
M(k) = M(k) - C_sw;
cooldown(k) = cooldown(k) - 1;
end
end
[best_M, best_k] = max(M);
if best_k ~= active_idx && (best_M - M(active_idx)) > delta
cooldown(best_k) = N_cool;
active_idx = best_k;
switch_hist(t_idx) = 1;
end
ZT_hist(t_idx) = compute_zt(bank(active_idx), T);
```

```
end
fprintf('Mean ZT = %.4f | Switches = %d\n', mean(ZT_hist), sum(switch_hist));
end
```